

```
In [ ]: ASSIGNMENT NO.2
Train a Decision Tree or Logistic Regression model on basic URL features (e.g.
number of subdomains) to classify phishing websites.
```

```
In [ ]: NAME : VISHAKHA VIKAS CHAVAN
ROLL NO : EN24107026
```

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_m
```

```
In [2]: df = pd.read_csv("PhiUSIIL_Phishing_URL_Dataset.csv")
```

```
In [3]: df
```

```
Out[3]:
```

	URL	URLLength	Domain	DomainLength
0	https://www.southbankmosaics.com	31	www.southbankmosaics.com	2
1	https://www.uni-mainz.de	23	www.uni-mainz.de	1
2	https://www.voicefmradio.co.uk	29	www.voicefmradio.co.uk	2
3	https://www.sfnmjournal.com	26	www.sfnmjournal.com	1
4	https://www.rewildingargentina.org	33	www.rewildingargentina.org	2
...	...	...	...	...
235790	https://www.skincareliving.com	29	www.skincareliving.com	2
235791	https://www.winchester.gov.uk	28	www.winchester.gov.uk	2
235792	https://www.nononsensedesign.be	30	www.nononsensedesign.be	2
235793	https://patient-cell-40f5.updatedlogmylogin.wo...	55	patient-cell-40f5.updatedlogmylogin.workers.dev	4
235794	https://www.alternativefinland.com	33	www.alternativefinland.com	2

235795 rows × 55 columns

```
In [4]: df.head()
```

Out [4]:

	URL	URLLength	Domain	DomainLength	IsDomain
0	https://www.southbankmosaics.com	31	www.southbankmosaics.com	24	
1	https://www.uni-mainz.de	23	www.uni-mainz.de	16	
2	https://www.voicefmradio.co.uk	29	www.voicefmradio.co.uk	22	
3	https://www.sfnmjournal.com	26	www.sfnmjournal.com	19	
4	https://www.rewildingargentina.org	33	www.rewildingargentina.org	26	

5 rows × 55 columns

In [5]: `df.tail()`

Out [5]:

	URL	URLLength	Domain	DomainLength	IsDomain
235790	https://www.skincareliving.com	29	www.skincareliving.com	22	
235791	https://www.winchester.gov.uk	28	www.winchester.gov.uk	21	
235792	https://www.nononsensedesign.be	30	www.nononsensedesign.be	23	
235793	https://patient-cell-40f5.updatedlogmylogin.wo...	55	patient-cell-40f5.updatedlogmylogin.workers.dev	47	
235794	https://www.alternativefinland.com	33	www.alternativefinland.com	26	

5 rows × 55 columns

In [6]: `df.isnull().sum()`

```

Out[6]: URL 0
        URLLength 0
        Domain 0
        DomainLength 0
        IsDomainIP 0
        TLD 0
        URLSimilarityIndex 0
        CharContinuationRate 0
        TLDLegitimateProb 0
        URLCharProb 0
        TLDLength 0
        NoOfSubDomain 0
        HasObfuscation 0
        NoOfObfuscatedChar 0
        ObfuscationRatio 0
        NoOfLettersInURL 0
        LetterRatioInURL 0
        NoOfDigitsInURL 0
        DigitRatioInURL 0
        NoOfEqualsInURL 0
        NoOfQMarkInURL 0
        NoOfAmpersandInURL 0
        NoOfOtherSpecialCharsInURL 0
        SpacialCharRatioInURL 0
        IsHTTPS 0
        LineOfCode 0
        LargestLineLength 0
        HasTitle 0
        Title 0
        DomainTitleMatchScore 0
        URLTitleMatchScore 0
        HasFavicon 0
        Robots 0
        IsResponsive 0
        NoOfURLRedirect 0
        NoOfSelfRedirect 0
        HasDescription 0
        NoOfPopup 0
        NoOfiFrame 0
        HasExternalFormSubmit 0
        HasSocialNet 0
        HasSubmitButton 0
        HasHiddenFields 0
        HasPasswordField 0
        Bank 0
        Pay 0
        Crypto 0
        HasCopyrightInfo 0
        NoOfImage 0
        NoOfCSS 0
        NoOfJS 0
        NoOfSelfRef 0
        NoOfEmptyRef 0
        NoOfExternalRef 0
        label 0
        dtype: int64

```

```
In [7]: df.columns
```

```
Out[7]: Index(['URL', 'URLLength', 'Domain', 'DomainLength', 'IsDomainIP', 'TLD',
            'URLSimilarityIndex', 'CharContinuationRate', 'TLDLegitimateProb',
            'URLCharProb', 'TLDLength', 'NoOfSubDomain', 'HasObfuscation',
            'NoOfObfuscatedChar', 'ObfuscationRatio', 'NoOfLettersInURL',
            'LetterRatioInURL', 'NoOfDegitsInURL', 'DegitRatioInURL',
            'NoOfEqualsInURL', 'NoOfQMarkInURL', 'NoOfAmpersandInURL',
            'NoOfOtherSpecialCharsInURL', 'SpacialCharRatioInURL', 'IsHTTPS',
            'LineOfCode', 'LargestLineLength', 'HasTitle', 'Title',
            'DomainTitleMatchScore', 'URLTitleMatchScore', 'HasFavicon', 'Robots',
            'IsResponsive', 'NoOfURLRedirect', 'NoOfSelfRedirect', 'HasDescriptio
n',
            'NoOfPopup', 'NoOfiFrame', 'HasExternalFormSubmit', 'HasSocialNet',
            'HasSubmitButton', 'HasHiddenFields', 'HasPasswordField', 'Bank', 'Pa
y',
            'Crypto', 'HasCopyrightInfo', 'NoOfImage', 'NoOfCSS', 'NoOfJS',
            'NoOfSelfRef', 'NoOfEmptyRef', 'NoOfExternalRef', 'label'],
            dtype='object')
```

```
In [9]: features = [
        'URLLength',
        'IsDomainIP',
        'NoOfSubDomain',
        'NoOfQMarkInURL',
        'NoOfEqualsInURL',
        'IsHTTPS'
    ]

X = df[features]
y = df['label']
```

```
In [10]: X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42, stratify=y
    )

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)

Training data: (188636, 6)
Testing data: (47159, 6)
```

```
In [11]: model = DecisionTreeClassifier(
        max_depth=5,
        random_state=42
    )

model.fit(X_train, y_train)

print("Model trained successfully!")

Model trained successfully!
```

```
In [12]: y_pred = model.predict(X_test)

print(y_pred[:20])

[1 1 1 1 1 0 1 1 1 0 1 0 0 0 1 1 1 1 1 0]
```

```
In [13]: accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

Accuracy: 0.9284972115608897

```
In [14]: print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.94	0.89	0.91	20189
1	0.92	0.96	0.94	26970
accuracy			0.93	47159
macro avg	0.93	0.92	0.93	47159
weighted avg	0.93	0.93	0.93	47159

```
In [15]: cm = confusion_matrix(y_test, y_pred)
```

```
print("Confusion Matrix:")
print(cm)
```

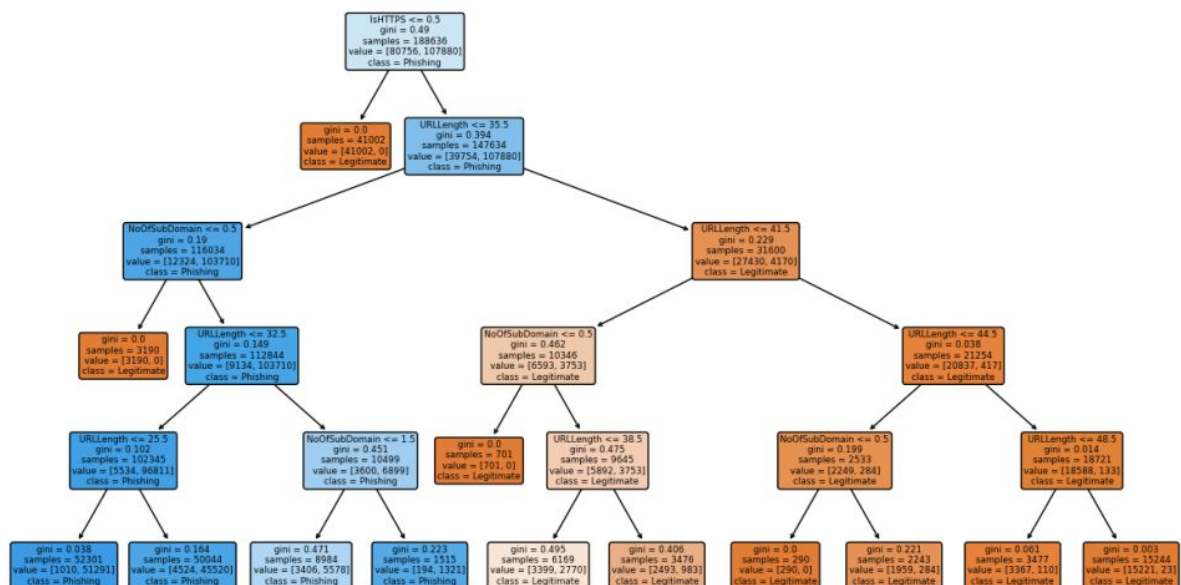
```
Confusion Matrix:
[[17869  2320]
 [ 1052 25918]]
```

```
In [16]: from sklearn.tree import plot_tree
```

```
plt.figure(figsize=(15, 8))

plot_tree(
    model,
    feature_names=features,
    class_names=["Legitimate", "Phishing"],
    filled=True,
    rounded=True
)

plt.show()
```



```
In [17]: print("Final Accuracy:", accuracy)
```

Final Accuracy: 0.9284972115608897